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PROFESSIONAL SUMMARY:

Dynamic and highly skilled Senior Data Engineer with a passion for building scalable and efficient data solutions. Seeking a challenging role where I can leverage my expertise in AWS data services and data engineering principles to design and implement robust data pipelines, optimize data storage, and enable data-driven decision-making. Committed to delivering high-quality solutions by leveraging my strong analytical skills, problem-solving abilities, and comprehensive understanding of data architecture.

- Over 10 years of professional IT experience, including 6+ years in cloud-based Big Data ecosystems with a strong emphasis on AWS data engineering services and 4+ years as a Data Warehouse/ETL developer.
- Experienced data professional specializing in designing, implementing, and managing end-to-end ETL/ELT pipelines on AWS and Informatica IICS, ensuring scalability, high availability, and performance optimization.
- Expert in AWS data services including S3 for raw/curated storage, Glue for ETL orchestration, Athena for serverless querying, Redshift for data warehousing, Kinesis Data Streams for streaming ingestion, and Lake Formation for centralized governance and access control.
- Designed and optimized large-scale data pipelines using AWS Glue, Lambda, Step Functions, EMR (Spark, Hive), Redshift Spectrum, and Informatica IICS for both batch and real-time processing.
- Proficient in integrating diverse on-premises and cloud data sources into AWS targets, implementing CDC (Change Data Capture) using Informatica IICS and AWS services, and building data models for analytical workloads.
- Developed streaming data solutions leveraging AWS Kinesis, Kafka, and Spark Streaming for low-latency analytics and real-time decision-making.
- Skilled in advanced SQL (DDL, DML), indexing, and query optimization for performance tuning in AWS Redshift, RDS, and Aurora environments.
- Applied data quality frameworks and validation checks across AWS pipelines to ensure accuracy, consistency, and compliance.
- Built CI/CD pipelines for AWS data engineering workloads using Jenkins and Code Pipeline, enabling automated testing and deployment.
- Collaborated with DevOps teams to monitor and troubleshoot pipelines, using AWS CloudWatch, CloudTrail, and Glue job metrics for proactive issue resolution.
- Designed data ingestion and transformation workflows using Apache Spark, Hive, HBase on AWS EMR, and Informatica IICS for high-volume processing
- Integrated curated datasets with BI platforms including Amazon Quick Sight, Power BI, and Tableau for dashboarding and analytics.
- Secondary expertise in Azure data services, including Azure Data Factory, Azure Databricks, Blob Storage, Logic Apps, and Azure Functions, to deliver multi-cloud interoperability and migration solutions.
- Experienced in Snowflake integration on AWS and Azure, implementing SCD, surrogate key generation, and optimized schema design for reporting workloads.
- Delivered multi-cloud data engineering solutions across AWS and Azure, ensuring seamless scalability, fault tolerance, and cost efficiency.
- Integrated AI/ML capabilities into large-scale data pipelines using PySpark MLlib, Pandas, TensorFlow, and PyTorch for feature engineering, model training, and real-time inference.
- Orchestrated machine learning workflows using Apache Airflow to automate data preprocessing, model retraining, and deployment in production environments.
- Developed automated data preparation workflows combining Pandas for in-memory processing, PySpark for distributed transformations, and Informatica IICS for enterprise-scale ETL, ensuring efficient readiness of datasets for AI/ML applications.
- Active participant in Agile methodologies using JIRA for sprint planning, daily stand-ups, and cross-team collaboration to deliver high-quality, production-ready solutions.

TECHNICAL SKILLS

AWS Services	S3, EC2, Glue, Redshift, Step Functions, RDS, Kinesis Data Stream, Code Pipeline, Cloud Watch, Quick Sight, SageMaker.
Azure Services	Azure data Factory, Airflow, Azure Data Bricks, Logic Apps, Functional App, Snowflake, Azure DevOps, Azure Synapse Analytics.
Big Data & AI/ML Technologies	MapReduce, Hive, Python, PySpark, PySpark MLlib, Scala, Kafka, Spark streaming, Oozie, Sqoop, Zookeeper, TensorFlow, PyTorch, Keras.
ETL & Integration Tools	Informatica IICS, Azure Data Factory, AWS Glue, Apache NiFi
Data Visualization	Power BI, Tableau, Quick Sight
Hadoop Distribution	Cloudera, Horton Works.
Languages	Java, SQL, PL/SQL, Python, HiveQL, Scala.
Operating Systems	Windows (XP/7/8/10), UNIX, LINUX, UBUNTU, CENTOS.
Build Automation tools	Ant, Maven
Version Control	GIT, GitHub.
IDE & Build Tools, Design	Eclipse, Visual Studio.
Databases	MS SQL Server 2016/2014/2012, Azure SQL DB, Azure Synapse. MS Excel, MS Access, Oracle 11g/12c, Cosmos DB

EDUCATION

- Bachelors in Electronics & Communication Engineering from Jawaharlal Nehru Technological University Hyderabad, India.
- Masters in Information Technology at University of North Texas

CERTIFICATION

- AWS Certified Data Engineer – Associate.
- Microsoft Certified - Azure Solution Architect.

WORK EXPERIENCE

Role: Sr. Data Engineer **Feb 2024 – Till Now**

Client: FM Global, Johnston, RI

Responsibilities:

- Managed end-to-end ETL/ELT data pipelines using AWS Glue, Informatica IICS, and AWS Step Functions**,** ensuring scalability, reliability, and smooth operations.
- Designed optimized query techniques and indexing strategies to enhance data retrieval in Amazon Redshift and Snowflake.
- Utilized advanced SQL (DDL, DML) and database objects (indexes, triggers, views, stored procedures, functions) for data manipulation in cloud and on-prem environments.
- Integrated on-premises data sources (MySQL, Cassandra) with Amazon S3, Amazon RDS, and Snowflake using AWS Glue, Informatica IICS, and AWS Database Migration Service (DMS**)**, applying transformations for analytics readiness.
- Orchestrated data movement into SQL databases using AWS Glue workflows and AWS Lambda.
- Applied data warehousing techniques including data cleansing, SCD handling, surrogate key assignment, and CDC for Snowflake modeling on AWS.
- Designed ingestion pipelines using Amazon Kinesis Data Streams**,** Apache NiFi, and Flume integrated with AWS for high-volume, real-time data processing.
- Developed and maintained ETL workflows with AWS Glue (PySpark), Informatica IICS, Amazon EMR (Spark, Hive), Apache Airflow, and Apache Beam.
- Applied ML libraries in production pipelines, including PyTorch and TensorFlow for predictive modeling, PySpark MLlib for large-scale machine learning workflows, and Pandas for feature engineering and preprocessing.
- Implemented automated data quality checks and cleansing frameworks in AWS Glue and EMR.
- Built optimized data models in Snowflake**,** Amazon Redshift**,** Hive, and HBase for efficient BI and reporting.
- Created transformations and validations using Spark-SQL in AWS EMR, integrating with Amazon Quick Sight dashboards.
- Automated workflows using AWS Lambda and Amazon Event Bridge for event-driven and scheduled data processing.
- Designed data integration across Hadoop, Amazon S3**,** and RDBMS environments.
- Implemented CI/CD pipelines for data workflows using Jenkins, AWS Code Pipeline, and Git.
- Collaborated with DevOps to integrate test-driven data pipelines in AWS Code Build.

- Developed Hive scripts for distributed processing with Hive on Spark in EMR.
- Delivered real-time streaming solutions with Kinesis, Spark Streaming, and Hive integrated with AWS analytics.
- Wrote high-performance Scala and Python code for big data in AWS EMR and Spark clusters.
- Used JIRA for Agile project management, participating in sprint planning and PI sessions.

Environment: AWS Glue, Amazon EMR, Informatica IICS**,** Amazon S3, Amazon Redshift, Amazon RDS, AWS Lambda, AWS Code Pipeline, AWS Code Build, AWS Step Functions, Amazon Kinesis, Snowflake, MS SQL, Oracle, HDFS, MapReduce, YARN, Spark, Hive, SQL, Python, Scala, PySpark, PySpark MLlib, Pandas, TensorFlow, PyTorch, Spark Performance, Data Integration, Data Modeling, Data Pipelines, Production Support, Shell Scripting, Git, JIRA, Jenkins, Apache NiFi, Flume, Quick Sight, Apache Airflow.

Role: Sr. Data Engineer **Nov 2021 – Dec 2023**

Client: Discover Financial, Riverwoods, IL

Responsibilities:

- Optimized large-scale data processing on AWS by enhancing Spark performance through partitioning, caching, and broadcast variables.
- Designed and developed scalable ETL/ELT pipelines using AWS Glue, Informatica IICS, Amazon EMR, and AWS Lambda to process structured and semi-structured data.
- Ingested data from Amazon S3, APIs, and on-premises databases, applying complex transformations with PySpark and Scala.
- Migrated enterprise Oracle and SQL Server workloads to Amazon Redshift, Amazon RDS, and Snowflake using AWS DMS, Informatica IICS, and Lake Formation for secure governance.
- Built real-time and streaming data ingestion frameworks leveraging Amazon Kinesis Data Streams**,** Apache Kafka, and Apache NiFi.
- Designed optimized data models in Redshift**,** Hive, and Parquet to support advanced analytics and reporting in Amazon Quick Sight.
- Utilized Redshift Spectrum and Amazon Athena for cost-effective querying of S3-based datasets.
- Automated data validation and cleansing processes within AWS Glue, EMR, and Informatica IICS pipelines.
- Orchestrated batch and streaming workflows using Apache Airflow integrated with AWS services.
- Applied AI/ML libraries such as PyTorch, TensorFlow, PySpark MLlib, and Pandas for predictive modeling, feature engineering, and large-scale machine learning workflows on AWS cloud datasets.
- Integrated ML model training, evaluation, and inference pipelines with Airflow and EMR, supporting automated, scalable, production-ready machine learning workflows.
- Implemented CI/CD pipelines for data workflows using AWS Code Pipeline, Code Build, Jenkins, and Informatica IICS deployment automation**.**
- Collaborated with cross-functional teams to architect cost-optimized, fault-tolerant AWS-based data solutions.
- Provided production support, addressing data quality issues, performance bottlenecks, and pipeline failures in **AWS and Informatica IICS** environments.
- Applied Agile methodologies using JIRA and maintained version control with **Git** and Jenkins.

Environment: AWS Glue, Amazon EMR, Informatica, S3, Redshift, RDS, Snowflake, DMS, Lake Formation, Kinesis, Lambda, Code Pipeline, Code Build, Athena, Spectrum, Quick Sight, Kafka, NiFi, Airflow, Python, Scala, Hive, Git, JIRA, Jenkins.

Role: Data Engineer **Mar 2019 – Sept 2021**

Client: State of Oregon(Dept. Of Health), Portland, OR

Responsibilities:

- Designed and implemented an Enterprise Data Lake using Amazon S3 to support diverse use cases, including analytics, large-scale processing, storage, and reporting of rapidly changing data.
- Maintained high-quality reference data by performing cleaning, transformation, and validation within Amazon RDS and Amazon Redshift, collaborating with stakeholders and solution architects.
- Developed tabular and semantic models in Amazon Quick Sight to meet enterprise reporting and visualization needs.
- Executed data ingestion pipelines into various AWS services (Amazon S3**,** Amazon RDS**,** Amazon Redshift**,** Snowflake) and processed data using Amazon EMR with PySpark.
- Created complex ETL workflows leveraging AWS Glue and PySpark on EMR for large-scale transformations.
- Managed data loading workflows from Amazon S3 into Amazon Redshift and Snowflake.
- Developed reusable Python**,** PySpark, and Bash scripts for efficient data transformation and loading across on-premises and AWS platforms.

- Utilized Apache Spark SQL and Streaming capabilities on EMR and Amazon Kinesis for intraday and real-time data processing.
- Configured Kerberos authentication for secure communication on EMR clusters, validating access for HDFS, Hive, and MapReduce users.
- Employed Spark SQL with Scala and Python, converting RDD case classes to schema RDDs for seamless distributed processing.
- Imported data from diverse sources such as HDFS and HBase into Spark RDDs and transformed it using PySpark to produce actionable datasets.
- Applied performance optimizations including distributed caching, partitioning, bucketing, and map-side joins in Hive on EMR.
- Developed reusable ETL frameworks automating migrations from RDBMS to Amazon S3 Data Lake using Spark Data Sources and Hive objects.
- Performed database import/export tasks with AWS DMS and traditional ETL tools like SSIS.
- Leveraged reporting tools including Amazon Quick Sight**,** Tableau, and SSRS for BI and visualization.

Environment: AWS Glue, Amazon EMR, Amazon S3, Amazon Redshift, Amazon RDS, Snowflake, AWS DMS, Kinesis, Hive, Spark, PySpark, Scala, HBase, Sqoop, Python, Bash, SSRS, Tableau, Quick Sight.

Role: Big data Developer **Jun 2018 – Jan 2020**
Client: AT&T, Dallas, TX
Responsibilities:

- Designed and developed scalable data transformation applications on Azure Data Lake Storage (ADLS) to enable efficient analytics for business stakeholders.
- Built and optimized data ingestion pipelines using Azure Data Factory (ADF) to ingest data from diverse sources into Azure Data Lake**,** Azure SQL Database, and Azure Synapse Analytics**.**
- Developed complex data flows and transformations leveraging PySpark within Azure Databricks, improving processing performance and scalability.
- Implemented batch and streaming data processing solutions using Apache Spark and Spark Streaming on Azure Databricks to support real-time analytics requirements.
- Created and managed schema designs and data models **in** Azure Synapse Analytics and Snowflake to support fast and reliable reporting and BI.
- Automated data workflows using Azure Logic Apps and Azure Functions for event-driven and scheduled processing.
- Leveraged Azure DevOps and Jenkins to establish CI/CD pipelines, ensuring smooth deployment of data engineering solutions.
- Integrated big data tools such as Apache Kafka**,** Hive**,** HBase, and Sqoop with Azure services to facilitate large-scale ingestion and processing.
- Utilized Shell scripting and Python for automating ETL workflows, managing system operations, and orchestrating pipelines.
- Ensured data quality and governance by implementing validation checks and cleansing frameworks within ADF and Databricks pipelines.
- Collaborated with cross-functional teams to design end-to-end Azure-based data architectures aligned with evolving business needs.

Environment: Azure Data Lake Storage, Azure Data Factory, Azure Synapse Analytics, Azure Databricks, Azure SQL Database, Snowflake, Azure Functions, Azure Logic Apps, Azure DevOps, Kafka, Hive, HBase, Sqoop, Hadoop, HDFS, Spark, MapReduce, Pig, Python, Shell Scripting, Jenkins.

Role: Data warehouse Developer **Apr 2016 – Apr 2018**
Client: Wells Fargo, Charlotte, NC
Responsibilities:

- Designed, created, and maintained databases for Server Inventory and Performance Inventory to support enterprise reporting requirements.
- Participated in Agile Scrum ceremonies; utilized Visual SourceSafe for version control and Trello for project tracking to ensure timely delivery.

- Developed drill-through and drill-down interactive reports with dynamic filters, sorting, and subtotal features using Power BI.
- Built Data Marts feeding downstream reports and developed a User Access Tool enabling users to create ad-hoc reports and query OLAP cubes.
- Created robust ETL packages using SSIS to extract data from heterogeneous sources, transform it, and load into the data warehouse.
- Automated report generation and cube refresh processes through SSIS jobs, improving data availability and workflow efficiency.
- Deployed SSIS packages to production, employing environment-independent configurations for flexibility across deployment stages.
- Utilized SQL Server Reporting Services (SSRS) to design, author, and deliver both interactive web-based and paper-based reports.
- Developed complex stored procedures and triggers to ensure consistent and accurate data entry within the databases.
- Facilitated secure external data sharing using Snowflake, enabling rapid data exchange without building additional pipelines.

Environment: Windows server, MS SQL Server 2014, SSIS, SSAS, SSRS, SQL Profiler, Power BI, C#, Performance Point Server, MS Office, SharePoint

Role: Data warehouse Developer **Mar 2014 – Feb 2016**
Client: United Airlines, Chicago, IL
Responsibilities:

- Experience in developing complex store procedures, efficient triggers, required functions, creating indexes and indexed views for performance.
- Developed complex stored procedures, triggers, functions, indexes, and indexed views to optimize SQL Server performance.
- Monitored and tuned SQL Server environments for improved query execution and system efficiency.
- Designed and implemented ETL data flows using SSIS, creating mappings and workflows to extract and transform data from SQL Server, Access, and Excel sources.
- Performed dimensional data modelling for Data Mart design, including fact and dimension tables, and managed Slowly Changing Dimensions (SCD).
- Managed error and event handling in ETL processes using precedence constraints, breakpoints, checkpoints, and logging.
- Built and deployed SSAS cubes and dimensions with various architectures; authored MDX scripts to enhance BI functionality.
- Applied star and snowflake schema designs for efficient data warehousing solutions.
- Created parameterized, ad hoc, dashboard, and scorecard reports in SSRS, utilizing drill-down, drill-through, and cascading features.
- Collaborated effectively within teams, demonstrating strong communication and leadership skills to deliver client-focused solutions.

Environment: MS SQL Server 2016, Visual Studio 2017/2019, SSIS, SharePoint, MS Access, Team Foundation Server, Git.